

Retail Banking Loan Portfolio & Risk Analytics

Production SQL-driven diagnostic analysis of 38,576 consumer loans (\$435.75M principal originations)

Dataset Scope: 38,576 Applications	Total Origination: \$435,757,075	Total Collections: \$473,070,933	Portfolio Health: 86.18% Prime / Good
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Core Portfolio Performance Indicators

HIGH-LEVEL EXECUTIVE METRICS

TOTAL APPLICATIONS 38,576 MTD (Dec): 4,314 MoM Growth: +6.9%	TOTAL FUNDED CAPITAL \$435.76M MTD: \$53.98M PMTD: \$47.75M (+13.0%)	CASH RECEIVED / COLLECTED \$473.07M MTD: \$58.07M Collection Ratio: 1.085x
WEIGHTED AVG INTEREST RATE 12.05% MTD: 12.36% PMTD: 11.94% (+42 bps)	AVERAGE DEBT-TO-INCOME (DTI) 13.33% MTD: 13.67% PMTD: 13.30% (+37 bps)	NET PORTFOLIO ROI MARGIN +8.56% Net Positive Spread: +\$37.31M

Underlying Production Query Logic

T-SQL CONSOLIDATED KPI ARCHITECTURE

ANALYTICS/KPI_PORTFOLIO_SUMMARY.SQL	ENGINE: SQL SERVER / POSTGRESQL
<pre>WITH MonthlyAggregations AS (SELECT COUNT(id) AS Total_Apps, SUM(loan_amount) AS Total_Funded, SUM(total_payment) AS Total_Collected, AVG(int_rate) * 100.0 AS Avg_Int_Rate, AVG(dti) * 100.0 AS Avg_DTI FROM bank_loan_data) SELECT *, (Total_Collected - Total_Funded) AS Net_Revenue_Spread FROM MonthlyAggregations; -- MTD Filters utilize: WHERE MONTH(issue_date) = 12 (December Baseline Analysis)</pre>	

Engineering Insight: Rather than executing multiple disparate aggregation runs, leveraging modern SQL Common Table Expressions (CTEs) minimizes scan cycles over the primary transaction ledger, maintaining near real-time auditability across both month-to-date (MTD) and previous-month-to-date (PMTD) benchmarks.

A critical underwriting benchmark is the loan risk bifurcated between performing accounts (Fully Paid & Current) and non-performing delinquent liquidations (Charged Off).

Good Loan Asset Class 86.18% of Total 33,243 Accounts Capital Disbursed: \$370,224,850 Capital Recovered: \$435,786,170 Net Realized Gain: +\$65,561,320 (+17.7%)	Charged Off / Default Class 13.82% of Total 5,333 Accounts Capital Disbursed: \$65,532,225 Recovery Payments: \$37,284,763 Net Write-Off Loss: -\$28,247,462 (-43.1%)
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Comprehensive Status Matrix & MTD Comparison

STATUS DRILL-DOWN

LOAN STATUS	VOLUME	FUNDED AMOUNT	COLLECTED AMOUNT	AVG RATE	AVG DTI	MTD FUNDED	MTD RECEIVED
Fully Paid	32,145	\$351,358,350	\$411,586,256	11.64%	13.17%	\$41,302,025	\$47,815,851
Charged Off	5,333	\$65,532,225	\$37,284,763	13.88%	14.00%	\$8,732,775	\$5,324,211
Current	1,098	\$18,866,500	\$24,199,914	15.10%	14.72%	\$3,946,625	\$4,934,318

ANALYTICS/RISK_SEGMENTATION.SQL

QUALITY RATIO ANALYTICS

```
SELECT
  loan_status,
  COUNT(id) AS LoanCount,
  SUM(total_payment) AS Total_Amount_Received,
  SUM(loan_amount) AS Total_Funded_Amount,
  ROUND(AVG(int_rate * 100), 2) AS Interest_Rate_Pct,
  ROUND(AVG(dti * 100), 2) AS DTI_Pct
FROM bank_loan_data
GROUP BY loan_status
ORDER BY LoanCount DESC;
```

Risk Observation: Charged off accounts carry an average interest rate of 13.88% vs 11.64% on fully paid accounts, illustrating an adverse selection curve where higher borrower pricing correlates directly with higher default probability and elevated DTI ratios (14.00%).

Temporal Origination Trajectory (Jan – Dec)

SEASONALITY & SCALING TRENDS

Systematic month-over-month expansion demonstrates strong quarterly ramping, moving from \$25.03M funded in January to a peak of \$53.98M in December.

M#	MONTH	TOTAL APPLICATIONS	FUNDED CAPITAL	COLLECTED TOTAL	RECOVERY INDEX
01	January	2,332	\$25,031,650	\$27,578,836	1.10x
02	February	2,279	\$24,647,825	\$27,717,745	1.12x
03	March	2,627	\$28,875,700	\$32,264,400	1.12x
04	April	2,755	\$29,800,800	\$32,495,533	1.09x
05	May	2,911	\$31,738,350	\$33,750,523	1.06x
06	June	3,184	\$34,161,475	\$36,164,533	1.06x
07	July	3,366	\$35,813,900	\$38,827,220	1.08x
08	August	3,441	\$38,149,600	\$42,682,218	1.12x
09	September	3,536	\$40,907,725	\$43,983,948	1.08x
10	October	3,796	\$44,893,800	\$49,399,567	1.10x
11	November	4,035	\$47,754,825	\$50,132,030	1.05x
12	December	4,314	\$53,981,425	\$58,074,380	1.08x

Maturity Structure & Tenor Distribution

36 VS 60 MONTH COMMITMENTS

36-Month Short Term

28,237 Loans (73.2%)

Funded: **\$273.04M** | Collected: **\$294.71M**
Average Ticket Size: **\$9,669**

60-Month Long Term

10,339 Loans (26.8%)

Funded: **\$162.72M** | Collected: **\$178.36M**
Average Ticket Size: **\$15,738**

ANALYTICS/TEMPORAL_TENOR_BREAKDOWN.SQL

WINDOW ORDERING PIPELINE

```
SELECT
    MONTH(issue_date) AS Month_Num,
    DATENAME(MONTH, issue_date) AS Month_Name,
    COUNT(id) AS Total_Loan_Applications,
    SUM(loan_amount) AS Total_Funded_Amount,
    SUM(total_payment) AS Total_Amount_Received
FROM bank_loan_data
GROUP BY MONTH(issue_date), DATENAME(MONTH, issue_date)
ORDER BY Month_Num ASC;
```

Loan Purpose & Borrower Intent Concentration

CAPITAL DEPLOYMENT USE CASES

LOAN PURPOSE	VOLUME	FUNDED PRINCIPAL	TOTAL RECEIPTS	% PORTFOLIO
Debt Consolidation	18,214	\$232,459,675	\$253,801,871	53.3%
Credit Card Refinance	4,998	\$58,885,175	\$65,214,084	13.5%
Other Miscellaneous	3,824	\$31,155,750	\$33,289,676	7.1%
Home Improvement	2,876	\$33,350,775	\$36,380,930	7.6%
Major Purchase	2,110	\$17,251,600	\$18,676,927	4.0%
Small Business Commercial	1,776	\$24,123,100	\$23,814,817	5.5%
Car Financing	1,497	\$10,223,575	\$11,324,914	2.3%
Wedding / Personal	928	\$9,225,800	\$10,266,856	2.1%
Medical Expenses	667	\$5,533,225	\$5,851,372	1.3%
Moving, Educational, House, Other	1,686	\$13,548,150	\$14,449,446	3.3%

Employment Tenure & Residential Collateral Profile

DEMOGRAPHIC STABILITY FACTORS

TOP EMPLOYMENT EXPERIENCE COHORTS

EXPERIENCE	APPS	FUNDED
10+ years	8,870	\$116.12M
< 1 year	4,575	\$44.21M
2 years	4,382	\$44.97M
3 years	4,088	\$43.94M
4–5 years	6,701	\$74.57M

HOME OWNERSHIP BREAKDOWN

OWNERSHIP	APPS	FUNDED
RENT	18,439	\$185.77M
MORTGAGE	17,198	\$219.33M
OWN	2,838	\$29.60M
OTHER / NONE	101	\$1.06M

Key Finding: 85% of total capital is secured by borrowers with Rent or Mortgage profiles, while senior workforce professionals (10+ years tenure) represent the largest single ticket borrower segment (\$116.12M funded).

Top Regional Exposures & Geographic Distribution

STATE-LEVEL CAPITAL ALLOCATIONS

RANK	STATE	TOTAL APPLICATIONS	FUNDED CAPITAL	TOTAL COLLECTIONS	CAPITAL RECOVERY %
#1	California (CA)	6,894	\$78,484,125	\$83,901,234	106.9%
#2	New York (NY)	3,701	\$42,077,050	\$46,108,181	109.6%
#3	Florida (FL)	2,773	\$30,046,125	\$31,601,905	105.2%
#4	New Jersey (NJ)	1,822	\$21,657,475	\$23,425,159	108.2%
#5	Illinois (IL)	1,486	\$17,124,225	\$18,875,941	110.2%
#6	Pennsylvania (PA)	1,482	\$15,826,525	\$17,462,908	110.3%
#7	Georgia (GA)	1,355	\$15,480,325	\$16,728,040	108.1%
#8	Massachusetts (MA)	1,310	\$15,051,000	\$16,676,279	110.8%
#9	Ohio (OH)	1,188	\$12,991,375	\$14,330,148	110.3%
#10	Maryland (MD)	1,027	\$11,911,400	\$12,985,170	109.0%

GitHub Repository Architecture & Execution Guide

PORTFOLIO PRESENTATION GUIDE

Recommended GitHub Directory Layout:

```
retail-banking-loan-risk-analysis/
├── sql/
│   ├── 01_ddl_and_schema_design.sql      # Table structure, primary keys & indexes
│   ├── 02_portfolio_kpi_summary.sql      # Executive MTD/PMTD aggregations
│   ├── 03_risk_segmentation_good_bad.sql # Performing vs Charged-off matrix
│   └── 04_demographic_slices.sql         # State, purpose, tenure & tenor analysis
├── docs/
│   └── portfolio_report.pdf              # This executive deliverable
└── README.md                            # Comprehensive problem statement & findings
```

ANALYTICS/INTERACTIVE_CREDIT_GRADE_FILTER.SQL

PARAMETER-DRIVEN ANALYTICS

```
-- Evaluates sub-tier risk characteristics for high-credit quality portfolios (Grade 'A')
SELECT
    purpose AS Loan_Purpose,
    COUNT(id) AS Grade_A_Applications,
    SUM(loan_amount) AS Grade_A_Funded_Principal,
    SUM(total_payment) AS Grade_A_Cash_Collected
FROM bank_loan_data
WHERE grade = 'A'
GROUP BY purpose
ORDER BY Grade_A_Applications DESC;
```